

Module 4: Supervised Learning
Submodule: 4.6 Gaussian Discriminant Analysis
Lecture: Maximum Likelihood Estimation (MLE)

XCS229 Machine Learning

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I. Introduction

In this lecture, we discuss how to learn parameters from data using the principle of **Maximum Likelihood** (ML).

II. Likelihood Function

The likelihood function, denoted as $L(\phi, \mu_0, \mu_1, \Sigma)$, describes how likely it is to observe the given data based on certain parameters. This function is central to the **Maximum Likelihood Estimation** (MLE) method.

$$L(\phi, \mu_0, \mu_1, \Sigma) = P\left(\left(x^{(1)}, y^{(1)}\right), \dots, \left(x^{(n)}, y^{(n)}\right); \phi, \mu_0, \mu_1, \Sigma\right).$$

This equation states that the likelihood L is equal to the probability of observing all data points $(x^{(i)}, y^{(i)})$ given the parameters $\phi, \mu_0, \mu_1, \Sigma$. Here, i represents the i^{th} example from a dataset of n examples.

$$L(\phi, \mu_0, \mu_1, \Sigma) = \prod_{i=1}^n P\left(\left(x^{(i)}, y^{(i)}\right); \phi, \mu_0, \mu_1, \Sigma\right).$$

Since we assume the data are drawn independently from each other, we can factor the joint probability into a product of individual probabilities. This means that the overall likelihood is the product of the likelihoods of each example.

$$L(\phi, \mu_0, \mu_1, \Sigma) = \prod_{i=1}^n P\left(x^{(i)} \mid y^{(i)}; \phi, \mu_0, \mu_1, \Sigma\right) P\left(y^{(i)}; \phi, \mu_0, \mu_1, \Sigma\right).$$

Using the chain rule of probability, we can expand the individual likelihood for $(x^{(i)}, y^{(i)})$ into the conditional probability of $x^{(i)}$ given $y^{(i)}$ multiplied by the probability of $y^{(i)}$ itself. The likelihood is thus expressed with respect to $y^{(i)}$ and $x^{(i)}$.

$$L(\phi, \mu_0, \mu_1, \Sigma) = \prod_{i=1}^n P\left(x^{(i)} \mid y^{(i)}; \mu_0, \mu_1, \Sigma\right) P\left(y^{(i)}; \phi\right).$$

Here we can simplify further by recognizing that $y^{(i)}$ only depends on the parameter ϕ and not μ_0, μ_1 or Σ . Therefore, the probability of $y^{(i)}$ is written solely as $P(y^{(i)}; \phi)$, while the conditional probability of $x^{(i)}$ given $y^{(i)}$ now depends on μ_0, μ_1, Σ alone.

Finally, in Maximum Likelihood Estimation, we seek to maximize the likelihood function:

$$\max_{\phi, \mu_0, \mu_1, \Sigma} L(\phi, \mu_0, \mu_1, \Sigma).$$

This leads us to find the most probable parameters $\phi, \mu_0, \mu_1, \Sigma$ that explain the observed data.

The independence assumption is common in machine learning because it often reflects the nature of data collection. However, dependencies can exist in certain scenarios (e.g., reinforcement learning).

III. Distinction Between Generative and Discriminative Learning Algorithms

In this section, we briefly highlight the key differences between generative and discriminative learning. Generative learning aims to model how the data is generated by learning the joint distribution of inputs x and outputs y . In contrast, discriminative learning focuses on modeling the decision boundary between classes by estimating the conditional probability of the output y given the input x . Understanding these differences is essential for choosing the appropriate model based on the specific problem we are addressing. Now, let's delve into generative learning first.

Generative Learning

- **Maximum Likelihood:**

- In generative algorithms, we maximize the joint probability of the input x and output y .
- Specifically, the aim is to maximize the joint probability density of the occurrences of both x and y .

- **Joint Probability:**

$$\prod_{i=1}^n P\left(x^{(i)} \mid y^{(i)}; \mu_0, \mu_1, \Sigma\right) P\left(y^{(i)}; \phi\right).$$

- This equation illustrates that we model the joint probability by multiplying the likelihood of $x^{(i)}$ given $y^{(i)}$ and the prior probability of $y^{(i)}$.

- **Assumptions:**

- Generative approaches impose assumptions on both x and y . This means that you model the entire data structure, including how both variables interact.

Discriminative Learning

- **Conditional Likelihood:**

- In contrast, discriminative algorithms focus on maximizing the conditional likelihood, or simply the likelihood of y given x .
- This is often referred to as modeling the output y directly conditioned on the input x and parameters θ .

- **Modeling Focus:**

- Discriminative models do not concern themselves with how x is generated. Instead, they target how y is derived from x with the assumption that x is a deterministic quantity that is observed.

- **Assumptions:**

- The discriminative approach imposes fewer assumptions since it only models y given x . This limitation may lead to a more direct focus on the decision boundary between classes without needing to consider the distribution of x .

- **Conditional Probability:**

$$P\left(y^{(1)}, \dots, y^{(n)} \mid x^{(1)}, \dots, x^{(n)}; \theta\right) = \prod_{i=1}^n P\left(y^{(i)} \mid x^{(i)}; \theta\right).$$

- This equation shows that the conditional likelihood is expressed as the product of the likelihoods of each $y^{(i)}$ given its corresponding $x^{(i)}$ under the parameters θ .

Summary of Differences

- **Focus:** Generative models deal with the full joint distribution of both x and y , while discriminative models focus solely on the boundary between different classes of y , given x .
- **Assumptions:** Generative models assume both input and output distributions; discriminative models assume only the output distribution conditioned on the input.
- **Trade-offs:**
 - With generative models, there's a risk of overfitting because they strive to learn more about the full data.
 - With discriminative models the challenge lies in capturing enough information about y based on x alone.

Practical Considerations

- Deciding to use a generative or discriminative approach often depends on domain-specific knowledge and the structure of the data. If underlying assumptions about data are accurate (e.g., assuming Gaussian distributions), a generative model may generalize well.
- Conversely, if assumptions do not hold, it could lead to errors, emphasizing the delicate balance in model selection.

IV. Maximizing the Likelihood

In this section, we discuss how to maximize the likelihood function $L(\phi, \mu_0, \mu_1, \Sigma)$. This is crucial for empirically estimating the parameters of our model.

The **likelihood function** is expressed as:

$$\arg \max_{\phi, \mu_0, \mu_1, \Sigma} L(\phi, \mu_0, \mu_1, \Sigma).$$

To simplify computation, we often use the **log-likelihood**:

$$\begin{aligned} \ell(\phi, \mu_0, \mu_1, \Sigma) &= \arg \max \log L(\phi, \mu_0, \mu_1, \Sigma) \\ &= \arg \max \sum_{i=1}^n \log p\left(x^{(i)}|y^{(i)}; \mu_0, \mu_1, \Sigma\right) + \log p\left(y^{(i)}; \phi\right) \end{aligned}$$

Taking the logarithm of the likelihood function transforms the product of probabilities into a sum, which is mathematically easier to handle. This transformation is important because it simplifies computations and makes the resulting values more manageable, especially when dealing with very small or very large numbers.

To find the parameters that maximize the log-likelihood, we use the fact from calculus that the **gradient** of a function is zero at its maximum:

$$\nabla \ell(\phi, \mu_0, \mu_1, \Sigma) = 0.$$

This implies solving the following system of equations:

$$\frac{\partial \ell}{\partial \phi} = 0, \quad \frac{\partial \ell}{\partial \mu_0} = 0, \quad \frac{\partial \ell}{\partial \mu_1} = 0, \quad \frac{\partial \ell}{\partial \Sigma} = 0.$$

To solve these equations, we need to set the partial derivatives with respect to each parameter to zero. (Note that in this course, the derivative of any parameter will have the same shape as the parameter. As such, $\partial \ell / \partial \phi \in \mathbb{R}$ and $\partial \ell / \partial \mu_0, \partial \ell / \partial \mu_1, \partial \ell / \partial \Sigma \in \mathbb{R}^d$.) This yields four equations that must be solved simultaneously. This analytical approach is often preferred over numerical optimization because it allows for direct computation of the maximizer without iterative algorithms.

The solutions for the parameters μ_0, μ_1, ϕ and Σ obtained from these equations will provide meaningful insights into the model. Understanding these parameters will help you interpret the underlying data distribution and how well the model fits the data.

Now that we have established the framework for maximizing the likelihood, we will proceed to derive specific solutions for the parameters and interpret their significance.

V. Maximum Likelihood Estimation (MLE) Solutions

In this section, we provide solutions for the parameters involved in our model based on the Maximum Likelihood Estimation (MLE) approach.

Let us define the following notations:

$$\begin{aligned} U_0 &= \{i : y^{(i)} = 0\} \quad (\text{indices for the negative examples}) \\ U_1 &= \{i : y^{(i)} = 1\} \quad (\text{indices for the positive examples}) \end{aligned}$$

The total number of examples is defined as:

$$n = |U_0| + |U_1|$$

where $|U_1|$ and $|U_0|$ denote the cardinality (size) of the sets of positive and negative examples, respectively.

The indicator function is defined as:

$$\mathbb{1}(E) = \begin{cases} 1 & \text{if } E \text{ happens,} \\ 0 & \text{otherwise.} \end{cases}$$

Estimating Parameter ϕ

The parameter ϕ , representing the fraction of positive examples, can be computed as:

$$\phi = \frac{|U_1|}{n}$$

This is the maximum likelihood estimate for the probability that $y = 1$.

Consequently, the fraction of negative examples is:

$$1 - \phi = \frac{|U_0|}{n}$$

Alternatively, we can express ϕ using the indicator function:

$$\phi = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{y^{(i)} = 1\}.$$

Estimating Parameters μ_0 and μ_1

Next, we estimate the means for the negative and positive examples as follows:

$$\mu_0 = \frac{1}{|U_0|} \sum_{i \in U_0} x^{(i)} = \frac{\sum_{i=1}^n \mathbb{1}\{y^{(i)} = 0\} \cdot x^{(i)}}{\sum_{i=1}^n \mathbb{1}\{y^{(i)} = 0\}}$$

This gives us the average of our negative examples.

Similarly, for the positive examples:

$$\mu_1 = \frac{1}{|U_1|} \sum_{i \in U_1} x^{(i)} = \frac{\sum_{i=1}^n \mathbb{1}\{y^{(i)} = 1\} \cdot x^{(i)}}{\sum_{i=1}^n \mathbb{1}\{y^{(i)} = 1\}}$$

This gives us the average of our positive examples.

Estimating Covariance Matrix Σ

The covariance matrix Σ can be estimated as:

$$\begin{aligned}\Sigma &= \frac{1}{n} \left(\sum_{i \in U_0} (x^{(i)} - \mu_0)(x^{(i)} - \mu_0)^T + \sum_{i \in U_1} (x^{(i)} - \mu_1)(x^{(i)} - \mu_1)^T \right) \\ &= \frac{1}{n} \sum_{i=1}^n \left(x^{(i)} - \mu_{y^{(i)}} \right) \left(x^{(i)} - \mu_{y^{(i)}} \right)^T \\ &= \frac{1}{n} \left(\sum_{i \in U_0} \left(x^{(i)} - \mu_0 \right) \left(x^{(i)} - \mu_0 \right)^T + \sum_{i \in U_1} \left(x^{(i)} - \mu_1 \right) \left(x^{(i)} - \mu_1 \right)^T \right).\end{aligned}$$

This formula provides the maximum likelihood estimate for the covariance matrix, accounting for both negative and positive examples.

At this point, we derived the MLE for all parameters ϕ, μ_0, μ_1 , and Σ . These parameters help us understand the distribution of our data.

VI. Summary

In this lecture, we covered essential concepts and methods related to **Maximum Likelihood Estimation** (MLE) within the context of Gaussian Discriminant Analysis. Here are the key takeaways:

- **Likelihood Function:** The likelihood function $L(\phi, \mu_0, \mu_1, \Sigma)$ measures the probability of observing the data given certain parameters. Our goal is to maximize this function to find the best-fit parameters for our model.
- **Log-Likelihood:** We often work with the log-likelihood $\ell(\phi, \mu_0, \mu_1, \Sigma)$, which simplifies the computation by transforming products into sums.
- **Parameter Estimation:**
 - The parameter ϕ represents the fraction of positive examples in the dataset.
 - The parameters μ_0 and μ_1 denote the means of the negative and positive classes, respectively.
 - The covariance matrix Σ captures the variability within the classes.
- **Generative vs. Discriminative Models:** Understanding the distinction between generative models (which model the joint distribution of x and y) and discriminative models (which model the conditional probability of y given x) is crucial in selecting the appropriate learning approach.

- **Practical Implications:** The choice between generative and discriminative models should be informed by the nature of the data and the assumptions you can reasonably make, which can greatly impact model performance and the validity of predictions.

These concepts form a foundational understanding of how to approach parameter estimation in Gaussian Discriminant Analysis using Maximum Likelihood Estimation. In the next lectures, we will discuss how to utilize these parameters for making predictions on new examples based on the learned model.